Customer 360 Data Integration

Bootcamp Project - 3

Project Objective

The objective of this project is to build a scalable, metadata-driven data ingestion framework that can dynamically ingest project data files from a GitHub repository and load them into Azure Data Lake Storage Gen2 (ADLS Gen2) using a single reusable pipeline.

The framework reduces the need to create separate pipelines for each source file and supports onboarding new files through configuration changes rather than code modifications.

Business Problem

Organizations often need to ingest multiple data files from a shared source repository. For this project, the source is the GitHub repository below, rather than REST APIs.

Source GitHub path: https://github.com/abhishekGam/ncpl_41/tree/main/Bootcamp/project3

Creating a separate pipeline for every file can lead to:

• High maintenance effort

• Duplicate code

• Increased deployment complexity

• Difficult onboarding of new files

A metadata-driven framework solves this by creating a generic file-ingestion pattern.

• Tool: Draw.io to design the architecture diagram.

Role

Assume the role of Data Engineers responsible for implementing an end-to-end data pipeline in Azure.

Tools and Technologies

• Azure Synapse Analytics

• Azure Data Lake Storage (ADLS Gen2)

• Azure SQL Database

• Serverless SQL pool

• Power BI

Step-by-Step Process

Step 1: Ingest Data

Use Azure Synapse Analytics pipelines to ingest project data files from the GitHub source repository into the raw container in ADLS Gen2.

Source: https://github.com/abhishekGam/ncpl_41/tree/main/Bootcamp/project3

Requirement: Dynamic GitHub File Ingestion Framework

Design and implement a metadata-driven ingestion framework to dynamically extract project data files from the GitHub repository using a reusable pipeline and load the data into Azure Data Lake Storage Gen2 (ADLS Gen2).

Destination: Azure SQL Database

Load data from ADLS Gen2 into Azure SQL Database using a scalable, metadata-driven framework, enabling dynamic loading of files from the data lake into target database tables through a reusable pipeline.

Project 2nd Part

Source: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Step 1: Download the Yellow Taxi Parquet files for January, February, March, and April and upload them into ADLS Gen2.

Step 2: Convert the Parquet files into a Delta table, partitioning the data by year and month.

Step 3: Create an external data source and a view on top of it.

Step 4: Feed the Delta table into Power BI reports.